EXPERIMENT – 8

AIM :

To implement banking-style transactions using BEGIN, COMMIT, ROLLBACK, and SAVEPOINT. To simulate partial transaction failures and demonstrate isolation levels and concurrency anomalies using two database sessions.

SOURCE CODE :

```
-- Create Bank Account Table

CREATE TABLE Bank (
   AccountID INT PRIMARY KEY,
   Name VARCHAR(50),
   Balance DECIMAL(10,2)
);

INSERT INTO Bank VALUES
(101, 'Amit', 5000),
(102, 'Ravi', 3000);

-- 1. Transaction Using COMMIT

START TRANSACTION;

UPDATE Bank
SET Balance = Balance - 1000
WHERE AccountID = 101;

UPDATE Bank
SET Balance = Balance + 1000
WHERE AccountID = 102;

COMMIT;

-- 2. Transaction Using ROLLBACK

START TRANSACTION;

UPDATE Bank
SET Balance = Balance - 500
WHERE AccountID = 101;

ROLLBACK;

-- 3. Transaction Using SAVEPOINT

START TRANSACTION;

UPDATE Bank
SET Balance = Balance - 500
WHERE AccountID = 101;

SAVEPOINT S1;
```

```
UPDATE Bank
SET Balance = Balance + 500
WHERE AccountID = 102;

ROLLBACK TO SAVEPOINT S1;

COMMIT;

-- 4. Display Final Balances

SELECT * FROM Bank;

-- 5. Demonstrate Isolation Level

SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;

START TRANSACTION;

SELECT * FROM Bank;

COMMIT;

-- 6. Demonstrate Concurrency
-- Execute in Session 1

START TRANSACTION;

UPDATE Bank
SET Balance = Balance - 100
WHERE AccountID = 101;

-- Execute in Session 2

SELECT * FROM Bank;

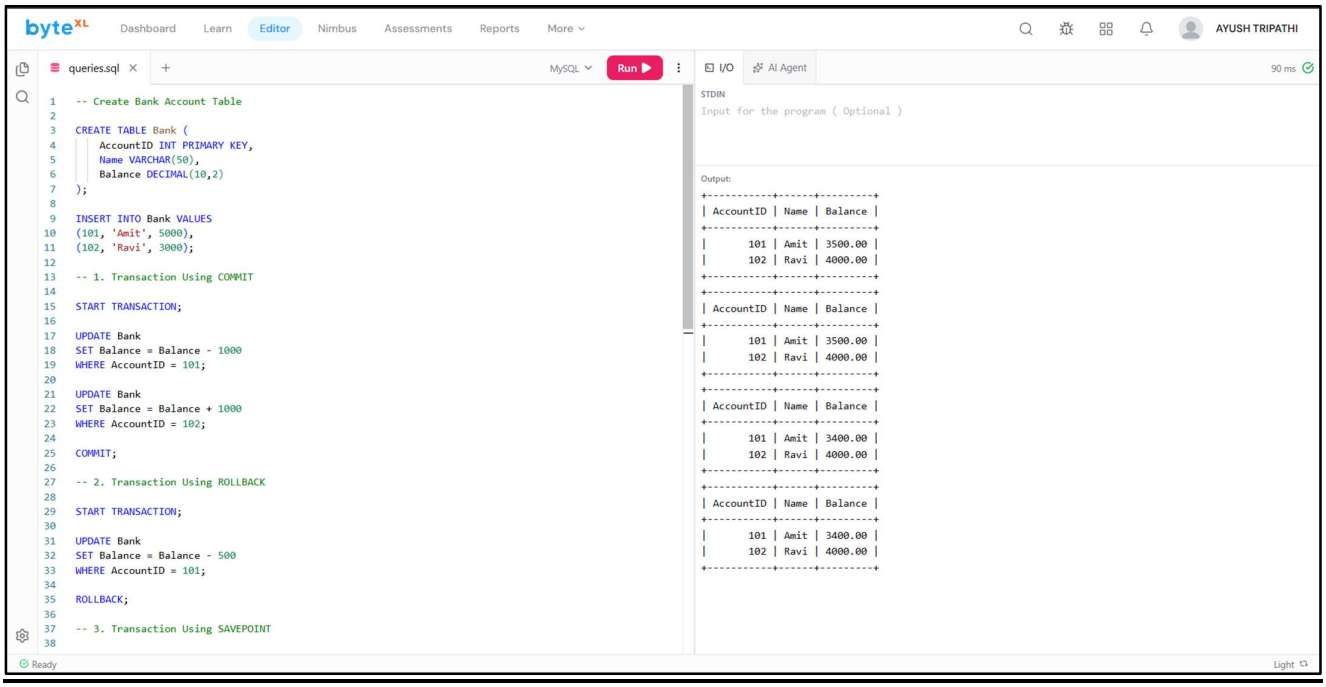
-- Return to Session 1

COMMIT;

-- Execute in Session 2 Again

SELECT * FROM Bank;
```

OUTPUT :


byteXL
Dashboard
Learn
Editor
Nimbus
Assessments
Reports
More
AYUSH TRIPATHI
queries.sql
MySQL
Run
I/O
AI Agent
90 ms
STDIN
Input for the program (Optional)
Output:
-- Create Bank Account Table
CREATE TABLE Bank (
AccountID INT PRIMARY KEY,
Name VARCHAR(50),
Balance DECIMAL(10,2)
);
INSERT INTO Bank VALUES
(101, 'Amit', 5000),
(102, 'Ravi', 3000);
-- 1. Transaction Using COMMIT
START TRANSACTION;
UPDATE Bank
SET Balance = Balance - 1000
WHERE AccountID = 101;
UPDATE Bank
SET Balance = Balance + 1000
WHERE AccountID = 102;
COMMIT;
-- 2. Transaction Using ROLLBACK
START TRANSACTION;
UPDATE Bank
SET Balance = Balance - 500
WHERE AccountID = 101;
ROLLBACK;
-- 3. Transaction Using SAVEPOINT
AccountID	Name	Balance
101	Amit	3500.00
102	Ravi	4000.00
AccountID	Name	Balance
101	Amit	3500.00
102	Ravi	4000.00
AccountID	Name	Balance
101	Amit	3400.00
102	Ravi	4000.00
AccountID	Name	Balance
101	Amit	3400.00
102	Ravi	4000.00
Ready
Light